

1. Administrative & Study Identification

Full Study Title: An Interventional Efficacy and Safety, Phase 3, Double-Blind, 2-Arm Study to Investigate Orally Administered Ibuzatrelevir Compared with Placebo in Non-Hospitalized Symptomatic Adult and Adolescent Participants with COVID-19 Who Are at High Risk of Progressing to Severe Illness **Short Title/ Acronym:** A Study to Learn About a Study Medicine Called Ibuzatrelevir in Adult and Adolescent Patients With COVID-19 Who Are Not Hospitalized But Are at Risk For Severe Disease **Protocol Number:** C5091017 (Other ID: 2024-517727-39-00) **NCT Number:** NCT06679140 **Sponsor Name:** Pfizer **Investigational Product:** Ibuzatrelevir (oral antiviral) **Phase of Development:** Phase 3 **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate whether ibuzatrelevir is effective and safe in adults and adolescents with COVID-19 who do not need to be in the hospital but who are at high risk for progression to severe disease, specifically by assessing the proportion of participants with COVID-19 related emergency department visits, all-cause hospitalization, and all-cause mortality. **Secondary Objectives:** To evaluate the time to sustained resolution of all COVID-19 targeted symptoms, the proportion of participants with post-acute COVID-19 medical events (including Long COVID symptoms), cardiovascular, renal, and pulmonary events, and changes from baseline in SARS-CoV-2 RNA levels.

2.2 Background & Rationale To address the need for a second-generation oral antiviral COVID-19 treatment for high-risk patients who are unable or unwilling to take nirmatrelvir/ritonavir (PAXLOVID). Because ibuzatrelevir can be taken as a single oral agent without ritonavir, it has the potential to eliminate ritonavir-associated drug-drug interactions and reduce adverse effects such as taste disturbances.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled (matching placebo) **Blinding:** Quadruple-blind (Double-blind, 2-arm study) **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target **\$\$\$:** 2,330 participants **Number of Sites:** United States + 18 other countries (Global)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. 12 to <18 years of age, weighing at least 40 kg, or ≥ 18 years of age of any weight at screening. 2. Presence of risk factors for progression to severe COVID-19 at the time of screening based on age (e.g., 12 to 49 years with ≥ 2 risk factors including moderate immunocompromise; ≥ 75 years with no requirement for additional risk factors). 3. Confirmed SARS-CoV-2 infection as determined by a Rapid Antigen Test (RAT) in a nasal or nasopharyngeal specimen collected within 1 day prior to randomization, with initial symptom onset within 5 days prior to randomization. 4. Unable or unwilling to take nirmatrelvir/ritonavir.

4.2 Exclusion Criteria 1. Current need or anticipated need for hospitalization within 24 hours due to signs of severe COVID-19 illness. 2. Receiving dialysis or having known severe renal impairment (eGFR consistently < 30 mL/min/1.73m²). 3. Has received any other antiviral for the treatment of COVID-19 (e.g., remdesivir, nirmatrelvir/ritonavir, molnupiravir) within 30 days or 5 half-lives prior to screening. 4. Current use of any prohibited concomitant medication(s).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ibuzatrelvir or matching placebo administered orally every 12 hours (twice daily). **Route of Administration:** Oral (Pill) **Duration of Treatment:** 5 days (10 doses total)

5.2 Concomitant & Prohibited Therapy Permitted: Co-administration of locally available standard of care for COVID-19 is allowed. **Prohibited:** Other antiviral medications for the treatment of COVID-19 or convalescent COVID-19 plasma.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening / Baseline	Day 1 Informed Consent, Medical History, Randomization, First Dose Interim
Day 1 to Day 28	Medication Administration (5 days), Symptom Tracking, Assessment of ED visits/Hospitalization	Follow-Up / End of Study	Up to 24 weeks Safety Labs, Efficacy Assessment for Long COVID, Telehealth/Onsite Visits (Approx. 12 appointments total)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Proportion of participants with COVID-19 related emergency department visits, all cause hospitalization, and all cause mortality. **Measurement Method:** Clinical assessment and tracking evaluated from Day 1 through Day 28.

7.2 Secondary & Exploratory Endpoints * Time to sustained resolution of all COVID-19 targeted symptoms. * Changes from baseline in SARS-CoV-2 RNA levels in nasopharyngeal/nasal swabs. * Proportion of participants with post-acute COVID-19 medical events and Long COVID symptoms.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via patient reporting and clinical assessment during the ~12 study appointments across the 24-week follow-up period. **Serious Adverse Events (SAEs):** Standard reporting timelines (typically 24 hours). **Data Monitoring Committee (DMC):** Yes, the trial utilizes an Independent Data Monitoring Committee.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcomes; Safety Set for AE/SAE evaluations. **Sample Size Justification:** $N = 2,330$, appropriately powered ($\alpha = 0.05$, $\text{Power} = 80\%$) to detect statistically significant reductions in COVID-19-related hospitalization or death. **Handling of Missing Data:** Standard multiple imputation/LOCF techniques applied for longitudinal symptom and viral RNA tracking.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Scheduled onsite and remote (telephone) monitoring visits throughout the 24-week study timeline. **Audit Trail:** eCRF locking and query resolution via standard trial quality audit mechanisms.

11. Study Identifiers & Governance

Primary ID: C5091017 **Registry Number:** NCT06679140 **Sponsor/Lead Organization:** Pfizer **Study Title:** A Study to Learn About a Study Medicine Called Ibuzatrelovir in Adult and Adolescent Patients

With COVID-19 Who Are Not Hospitalized But Are at Risk For Severe Disease **Official Regulatory Title:** An Interventional Efficacy and Safety, Phase 3, Double-Blind, 2-Arm Study to Investigate Orally Administered Ibuzatrelvir Compared with Placebo in Non-Hospitalized Symptomatic Adult and Adolescent Participants with COVID-19 Who Are at High Risk of Progressing to Severe Illness

12. Clinical Framework (COVID-19 Specific)

Primary Condition: COVID-19 / SARS-CoV-2 Infection **Diagnosis Threshold:** Confirmed SARS-CoV-2 infection (RAT+) with symptom onset within 5 days **Medical Methodology:** Antiviral Therapy (Ibuzatrelvir) **Optimization Target:** Prevention of severe disease progression, hospitalization, and mortality

13. Study Procedures & Methodology

Management Pathway: 5-day oral antiviral therapeutic course **Education Protocol (HCP):** Site initiation, protocol, and blinding compliance training **Education Protocol (Patient):** Dosing compliance (twice daily for 5 days) and self-monitoring tracking instructions **Quality Audit Mechanism:** Tracking of treatment adherence, viral RNA swab compliance, and endpoint event verifications

14. Observation & Follow-Up Schedule

Total Duration: 24 Weeks (approx. 6 months) **Onsite Visit Cadence:** Total of roughly 12 appointments (a mix of clinical onsite and telephone follow-ups) **Remote Monitoring Frequency:** Telephone visits included in the 12-appointment schedule **Checklist Review Interval:** Symptom diaries and monitoring from Day 1 to Day 28; extended follow-up for Long COVID events up to 24 weeks

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				